Digital Signature Standard (DSS)

Module V

DSS Introduction

- The NIST has published Federal Information Processing Standard (FIPS) – 186, known as the Digital Signature Standard (DSS).
- DSS utilizes SHA.
- DSS was originally proposed in 1991 and revised in 1993 in response to public feedback concerning the security of the scheme.

DSS steps

- Generation of public and private keys for USER A.
- Creation of digital signature by USER A for the message M.
- USER B verifying the digital signature.

DSS

Global public key components
{ p, q, g}. Sender public key y

Signature { r, s}

HACKER

Global public key components
{ p, q, g}. Sender public key y
and sender private key x

Global public key components
{ p, q, g}. Sender public key y

Message + signature: M||{r , s}

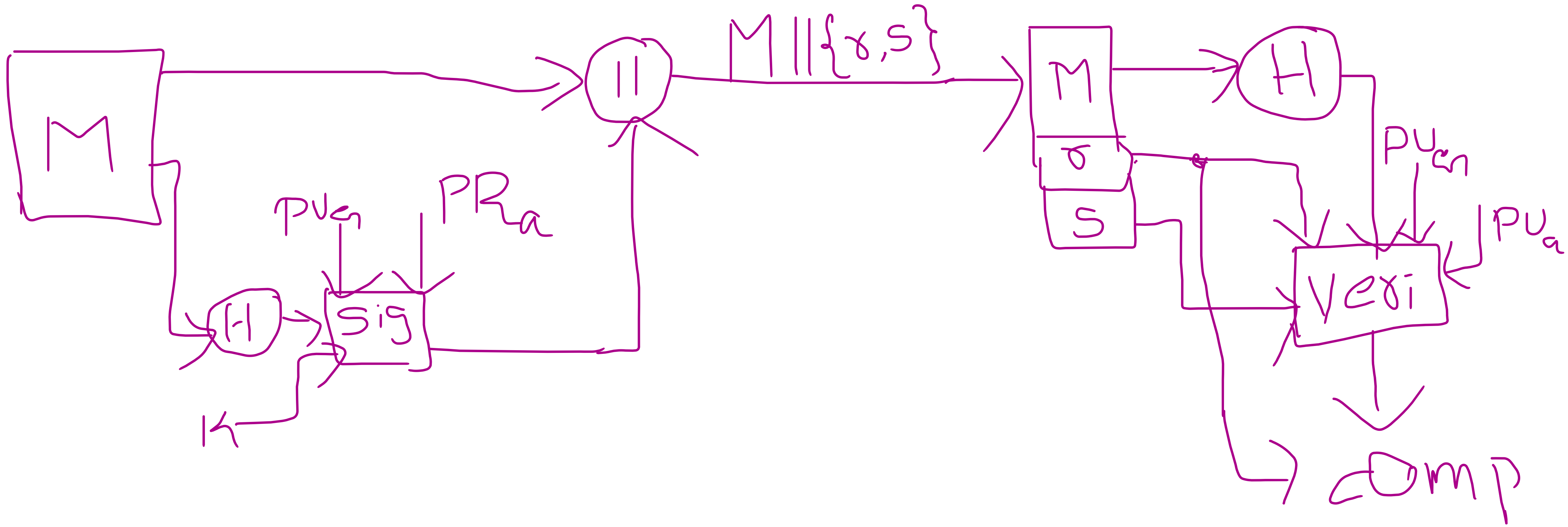
M||{r , s} ← signature

Insecure Channel

USER A

USER B

DSS cntd..



Generation of global key components {p, q, g}

- p: prime number where $2^{L-1} < p < 2^{L}$ for $512 \leq L \leq 1024$ and L a multiple of 64; i.e., bit length of between 512 and 1024 bits in increments of 64 bits.

  [512, 576, 640, 704, 768, 832, 896, 960, 1024]

- q: prime divisor of (p-1), where $2^{159} < q < 2^{160}$, i.e., bit length of 160.

- $g = h^{\frac{p-1}{q}} \, mod \, p$, where h is any Integer with $1 < h < p - 1$ such that $h^{\frac{p-1}{q}} \, mod \, p > 1$

Key pair generation

private key

x random or pseudorandom integer with $0 < x < q$

public key

$$y = g^x mod\ p$$

Signing

$$r = (g^k \bmod p) \bmod q$$

$$s = [k^{-1}(H(M) + x.r)] \bmod q$$

$$signature\ (r, s)$$

User's per message secret number

k=random/pseudorandom integer with 0<k<q

Verifying signature

$$w = (s')mod\ q)$$

$$u1 = [H(M')w]mod\ q$$

$$u2 = r'w\ mod\ q$$

$$\mathrm{v} = [\ (g^{u1}\ y^{u2})\ mod\ p]\ mod\ q$$

TEST v=r'

(r', s', m' received components)